

MAT 314 HW06

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Problem. 1.2. Let V be an abelian group. Prove that if V has a structure of $\mathbb{Q}$ -module with its given law of composition as addition, then that structure is uniquely determined.

Solution. We wish to show that there is only one map $\mathbb{Q} \times V \rightarrow V$ given by $(q, x) \mapsto qx$ such that $1x = x$, $q(x + y) = qx + qy$, $(q + r)x = qx + rx$, and $q(rx) = qr(x)$ for $q, r \in \mathbb{Q}$ and $x, y \in V$.

We have established that there is only one such map $\mathbb{Z} \times V \rightarrow V$ (i.e. abelian groups have exactly the natural $\mathbb{Z}$ -module structure), so we want to show that this structure can only be extended to that of a $\mathbb{Q}$ -module in at most one way. We can see that for $p/q \in \mathbb{Q}$ with $p, q \in \mathbb{Z}$ and $q > 0$,

$$\underbrace{\frac{p}{q}x + \cdots + \frac{p}{q}x}_{q \text{ times}} = q \cdot \frac{p}{q}x = px,$$

so $\frac{1}{q}x$ is an element of V that, when added to itself q times, equals x . If there are two such elements y and y' , then $q(y - y') = x - x = 0$, so multiplying by $\frac{1}{q}$ shows that $y = y'$. Thus there can only be at most one choice of $\frac{1}{q}x$, and then $\frac{p}{q}x$ is uniquely determined: it is either 0 (if $p = 0$), $\frac{1}{q}x$ added to itself p times (if $p > 0$), or $-\frac{1}{q}x$ added to itself p times (if $p < 0$). $\square$

Problem. 2.1. Let $R = \mathbb{C}[x, y]$, and let M be the ideal of R generated by the two elements x and y . Is M a free R -module?

Solution. Let $\{P_1, \dots, P_n\}$ be a generating set for M . We note that n cannot equal 1 because the ideal generated by $P \in R$ has every element divisible by P , which implies $P \mid x$ and $P \mid y$ so P is a constant polynomial, which is impossible because clearly M does not contain any polynomials with nonzero constant terms.

Now we show that $P_1, \dots, P_n$ are not independent: indeed,

$$-P_2 \cdot P_1 + P_1 \cdot P_2 + 0 \cdot P_3 + \cdots + 0 \cdot P_n = 0$$

but the coefficients of the P_i are not all zero. Therefore no generating set of M is a basis for M , so M cannot be a free R -module. $\square$

Problem. 3.3. Let A and B be $m \times m$ and $n \times n$ matrices, respectively. Prove that the trace of the linear operator $f(M) = AMB$ on the space $R^{m \times n}$ is the product $(\text{trace } A)(\text{trace } B)$.

Solution. Let $E_{i,j}$ be the $m \times n$ matrix whose only nonzero entry is a 1 in row i , column j . Clearly the $E_{i,j}$ are a basis for $R^{m \times n}$. The trace of f can be computed as

$$\sum_{i=1}^m \sum_{j=1}^n \langle E_{i,j}, f(E_{i,j}) \rangle$$

where $\langle X, Y \rangle = \sum_{i,j} X_{i,j} Y_{i,j}$ for $X, Y \in R^{m \times n}$. The summand is therefore the row i , column j entry of AMB . Now we can see easily that MB is the $m \times n$ matrix whose rows are all zeroes except for its i th row being equal to the j th row of B , and then the entry of $A(MB)$ in row i , column j is the product $a_{i,i} \cdot b_{j,j}$, since column j of MB has $b_{j,j}$ as its only nonzero entry, appearing in row i , and then the i th element in row i of A is $a_{i,i}$. Therefore the trace is equal to

$$\sum_{i=1}^m \sum_{j=1}^n a_{i,i} b_{j,j} = \left(\sum_{i=1}^m a_{i,i} \right) \left(\sum_{j=1}^n a_{j,j} \right).$$

(The question asked to use the permanence of identities thing, but this last equation just directly holds over arbitrary commutative rings so I don't really need to appeal to $\mathbb{C}$ here?) $\square$

Problem. 4.6. Let $\varphi: \mathbb{Z}^k \rightarrow \mathbb{Z}^k$ be a homomorphism given by multiplication by an integer matrix A . Show that the image of φ is of finite index if and only if A is nonsingular and that if so, then the index is equal to $|\det A|$.

Solution. We wish to show that $\mathbb{Z}^k/A\mathbb{Z}^k$ is finite with order $|\det A|$ if and only if A is nonsingular. We can write $A = Q^{-1}A'P$ for invertible P, Q where A' is diagonal, so that $\mathbb{Z}^k/A\mathbb{Z}^k \cong \mathbb{Z}^k/A'\mathbb{Z}^k$. Let the nonzero entries of the main diagonal of A' be positive integers $d_1, \dots, d_r$ for $r \leq k$ (i.e. Smith normal form, but we don't care about the divisibility information here). We can see that $A'\mathbb{Z}^k = d_1\mathbb{Z} \oplus \dots \oplus d_r\mathbb{Z}$, so if A (hence A') is nonsingular, then $r = k$, meaning

$$\mathbb{Z}^k/A'\mathbb{Z}^k \cong \bigoplus_{i=1}^k \mathbb{Z}/d_i\mathbb{Z},$$

and then the order of this group is $d_1 \cdots d_k = \det A' = |\det A|$. Conversely if A (hence A') is singular, then $r < k$, so

$$\mathbb{Z}^k/A'\mathbb{Z}^k \cong \mathbb{Z}^{k-r} \oplus \mathbb{Z}/d_1\mathbb{Z} \oplus \dots \oplus \mathbb{Z}/d_r\mathbb{Z}$$

which is infinite. $\square$

Problem. 7.4. In each case, identify the abelian group that has the given presentation matrix:

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 5 \end{pmatrix}, (2 \ 0 \ 0), \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 2 & 4 \\ 1 & 4 \end{pmatrix}, \begin{pmatrix} 2 & 4 \\ 6 & 4 \end{pmatrix}, \begin{pmatrix} 4 & 6 \\ 2 & 3 \end{pmatrix}.$$

Solution. In general for an $m \times n$ matrix A , the abelian group with presentation matrix A is $\mathbb{Z}^m/A\mathbb{Z}^n$.

- The first group is

$$\mathbb{Z}^2 / \begin{pmatrix} 2 \\ 1 \end{pmatrix} \mathbb{Z} = \mathbb{Z}^2 / \{(2x, x) : x \in \mathbb{Z}\} \cong \mathbb{Z},$$

noting that any element of $\mathbb{Z}^2$ can be written as $(a + 2x, x)$ for integers a, x .

- The second group is $\mathbb{Z}^2 / \{(0, 5x) : x \in \mathbb{Z}\} \cong \mathbb{Z} \oplus \mathbb{Z}/5\mathbb{Z}$.
- The third group is $\mathbb{Z}/(2 \ 0 \ 0) \cdot \mathbb{Z}^3 = \mathbb{Z}/2\mathbb{Z}$.
- The fourth group is the trivial group because $A\mathbb{Z}^3 = \mathbb{Z}^2$, as $A(x, y, 0) = (x, y)$ for any $(x, y) \in \mathbb{Z}^2$.
- The fifth group is also the trivial group because $\det A = 1$, hence it is an automorphism of $\mathbb{Z}^2$, i.e. $A\mathbb{Z}^2 = \mathbb{Z}^2$.
- After applying a change of generators and relations, we diagonalize this matrix as $\begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$, so the corresponding group is $\mathbb{Z}^2/(\mathbb{Z} \oplus 4\mathbb{Z}) = \mathbb{Z}/4\mathbb{Z}$.
- Similar to the previous group, the Smith normal form here is $\begin{pmatrix} 2 & 0 \\ 0 & 8 \end{pmatrix}$, so the quotient is $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/8\mathbb{Z}$.
- Finally, for the eighth group we can compute that $A\mathbb{Z}^2 = \{(2x, x) : x \in \mathbb{Z}\}$, so the quotient $\mathbb{Z}^2/A\mathbb{Z}^2$ is the same as the first group, namely $\mathbb{Z}$.

$\square$